

[Corresponding author]
[Affiliation, address]
[e-mail]
[Date]

The Editor
Nature Computational Science

Dear Editor,

We submit our Article “A resonance phase locates unstable self-similar singularities beyond an exponential precision wall” for consideration in *Nature Computational Science*.

The computational problem. Unstable self-similar blow-up profiles of incompressible flow are now being computed with physics-informed neural networks and classical solvers, and computer-assisted proofs need them to high accuracy. They come in hierarchies. How deep such a hierarchy can be computed, and how certain we can be that a computed profile is the one it is claimed to be, has not been asked. We show that the answer is limited by an exponential precision wall: each profile is a zero of a smoothness defect that shrinks about tenfold per profile. A solver’s residual does not reveal this. Our profiles have residuals of 10^{-13} , but whether they are profiles at all is decided at 10^{-9} .

What the paper shows, in incompressible porous-media flow:

1. **The anatomy of the wall.** A complete error audit of a converged solver at the deepest profiles finds three errors, none small or smooth:
 - a front-grid locking error, which we measure and cancel by grid shifting (it scales as $h_s^{7.1}$);
 - an angular-resolution error that stops converging;
 - round-off from deep origin truncation.An independent solver, which shares no numerical ingredient with the first, agrees to the resulting floor at every depth, while the positions of the profiles drift apart by five orders of magnitude.
2. **A pre-registered negative result.** Five predictions for the seventh profile and a decision rule were time-stamped in a public repository before the computation. The defect could not decide between them ($z_7 = 7.35 \pm 0.08$ against a decisiveness threshold of 0.003), and we report the test as not decisive, as the rule requires.
3. **An observable that bypasses the wall.** The resonance phase is a quadrature over $O(1)$ -accurate profile data. It moves 10 to more than 1,000 times less than the defect under the same numerical changes, and it locates the seventh profile more than ten times more precisely. Its imaginary part predicts the wall, so the precision that a direct method needs can be computed in advance.
4. **A validation lesson.** A failure of the phase computation passed every residual check. It was exposed by a geometric invariant and repaired with an exact scaling of the local problem, and we document it in full.
5. **A stated limit and a benchmark.** We leave the endpoint of the hierarchy open. We register three deeper profiles as a benchmark for neural and classical solvers, with their bands and the precision a decisive test needs.

Why *Nature Computational Science*. The paper turns a recurring difficulty of scientific computing, exponentially small quantities that decide qualitative structure, into an explicit lesson with a constructive remedy. Its practices are cheap and transfer to any search for unstable self-similar solutions, whether neural or classical:

- pre-registration with fixed decision rules;
- grid-shift measurement of periodic discretization errors;
- validation by geometric invariants rather than residuals;
- observables chosen for their insensitivity to the dominant errors.

We expect the work to interest computational physicists, numerical analysts, researchers in scientific machine learning, and mathematicians who use computer-assisted proofs.

Reproducibility. All code (MIT licence), the converged profiles and every logged output are openly available, together with the complete pre-registration record, including its corrections. Every figure and table is regenerated from the logs by one script each, and smoke tests run in seconds. A Code Ocean capsule will be provided for code review.

The manuscript is not under consideration elsewhere. [Either: A preprint has been posted on arXiv, [identifier]. Or: No preprint has been posted.] The authors declare no competing interests. The use of AI tools is described in the Methods, as required by the journal's policy.

Sincerely,

[Corresponding author], on behalf of all authors